SOC-5811 Week 5: Sampling distributions and statistical inference

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Review

Why are we creating models (e.g., linear regression) for data at all?

- Summarizing relationships.
- Making predictions.

Review

1. Choose a functional form for the model.

$$pcthouse_i = \beta_0 + \beta_1 pctpop_i + \epsilon_i$$

Once we define a model, how do we estimate the parameters using data?

Review

1. Choose a functional form for the model.

$$pcthouse_i = \beta_0 + \beta_1 pctpop_i + \epsilon_i$$

2. Choose an estimator (i.e., objective function) that relates the model to real observed data (e.g., OLS).

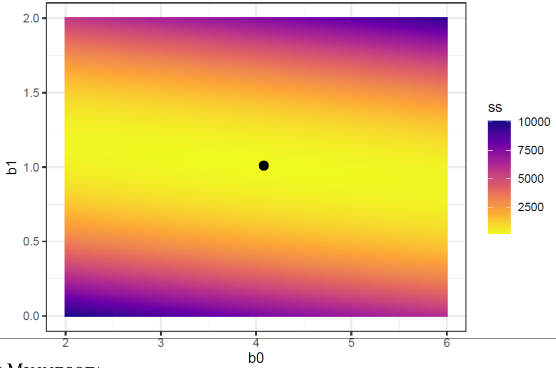
$$\text{Sum of squared residuals} = \sum_{i=1}^{n} (y_i - \widehat{\beta_0} - \widehat{\beta_1} x_i)^2$$

3. Fit the model (e.g., estimate parameters) by optimizing the objective function.

Review

What is happening behind the scenes to fit the model?

- If there is a closed-form solution: math.
- If there is not a closed-form solution: iterative calculations and approximations.



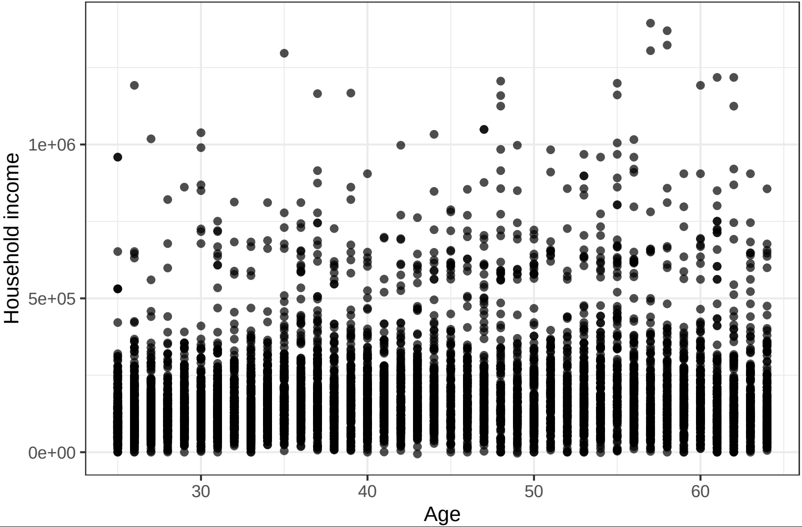
Review

- Linear regression is practically very useful because it summarizes relationships with simple parameters that are easy to interpret.
- This is true even once we start adding *more* independent variables!

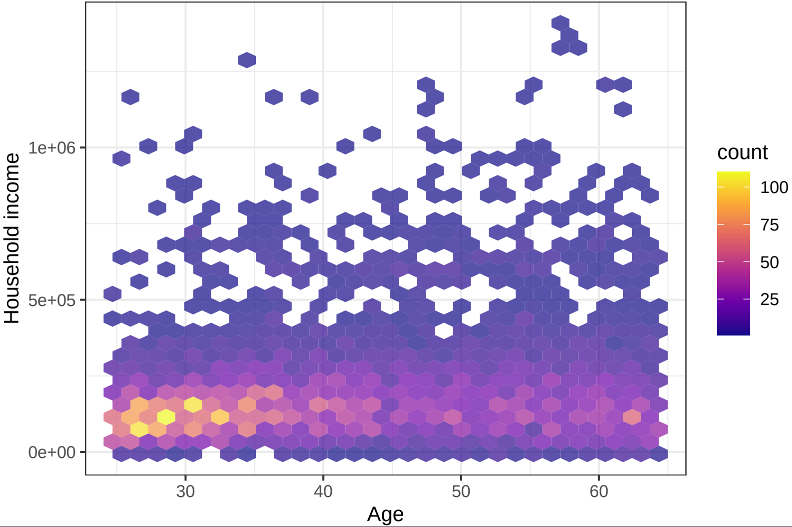
Statistical inference

- Across working years, are incomes higher at older ages in Hennepin County?
- If we can agree on definitions of income and age, then this seems like a question we should be able to answer empirically!

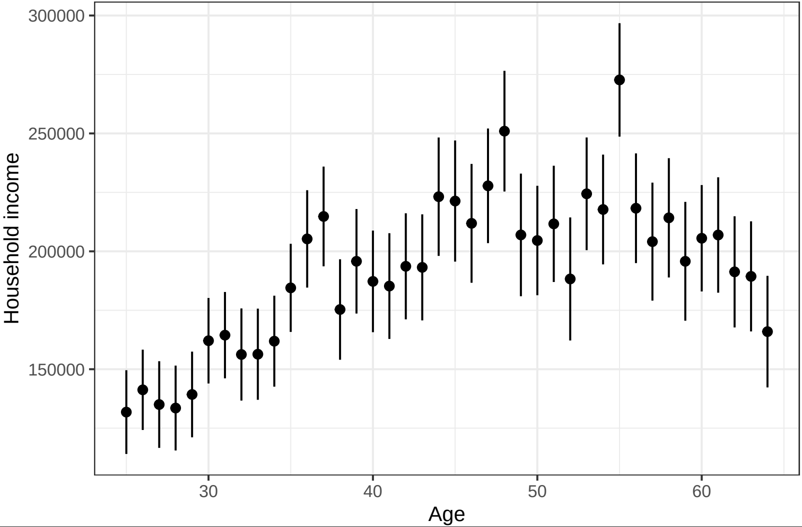
Statistical inference



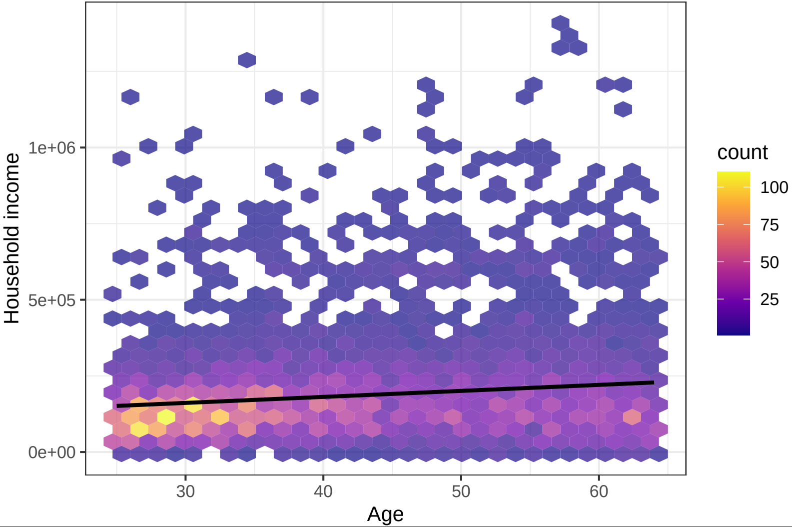
Statistical inference



Statistical inference



Statistical inference



Statistical inference

- Could the sample relationship be due to chance?

Statistical inference

- Could the sample relationship be due to chance?
- Based on our *sample*, how sure are we that the *population* parameter is different from 0?

Statistical inference

- **Parameters** = the unknown numbers that determine a statistical model.

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- Parameters can be used to simulate new data from the model.

Statistical inference

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- More simply, we observe data and we try to learn something about where it came from (i.e., the data generating process).

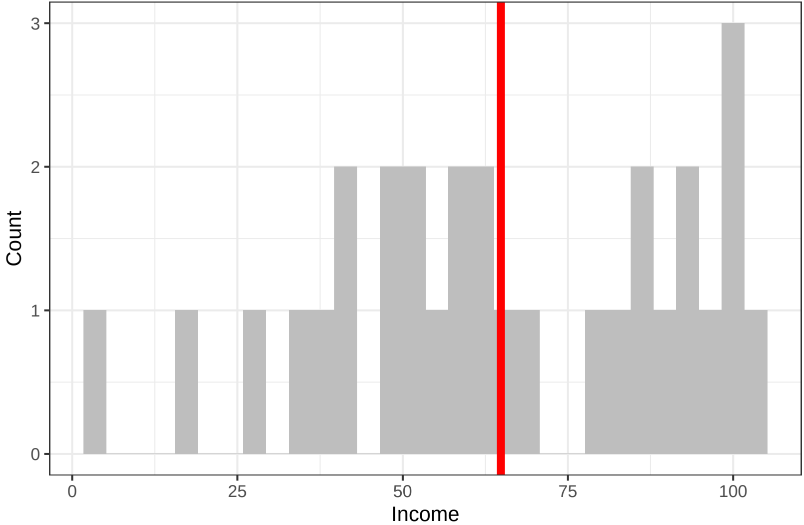
Statistical inference

- **Statistical inference** = a set of operations on data that yield estimates and uncertainty statements about **predictions and parameters** of some underlying process or population.
- More simply, we observe data and we try to learn something about where it came from (i.e., the data generating process).
- **This involves uncertainty.**

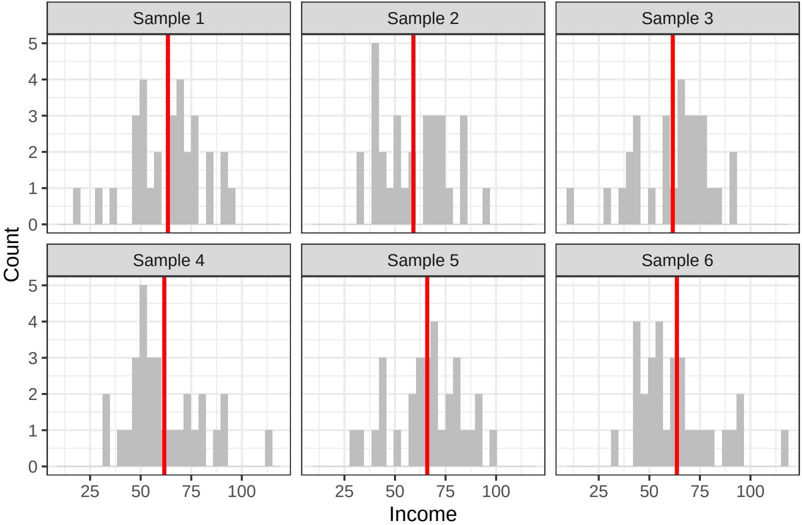
Statistical inference

- Say I have a population of 1000 people and I ask 30 of them their income. The parameter I'm interested in inferring something about is the average income in the entire population.

Statistical inference



Statistical inference



SAMPLING DISTRIBUTION

- **Sampling distribution** = the set of possible datasets that could have been observed if the data collection process had been re-done, along with the probabilities of these values.

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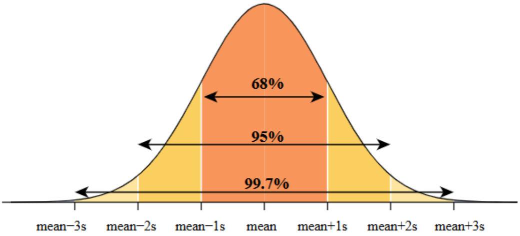
- In practice, we will not know the sampling distribution; we can only estimate it.

Standard errors

- **Standard deviation** = a measure of the spread of data around the mean.

$$\sigma = \sqrt{\frac{\sum(x_i - \mu)^2}{N}}$$



STANDARD ERRORS

- **Standard deviation** = a measure of the spread of data around the mean.

$$\sigma = \sqrt{\frac{\sum(x_i - \mu)^2}{N}}$$

- **Standard error** = ???

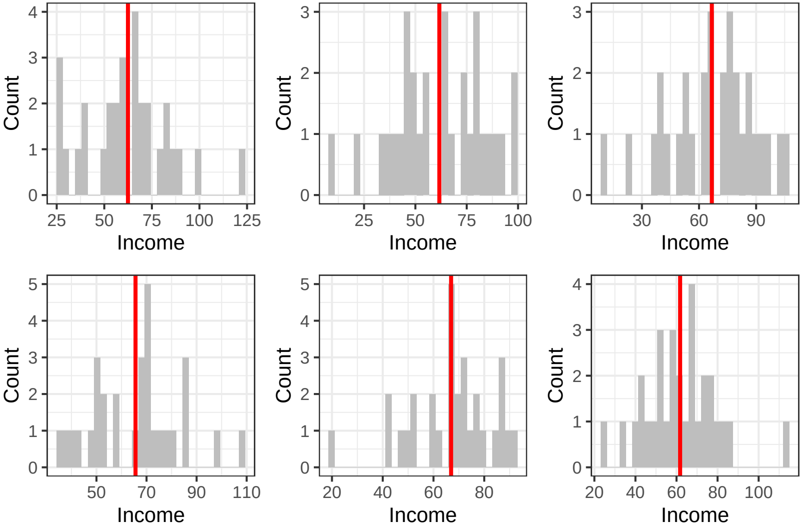
Standard errors

- **Standard deviation** = a measure of the spread of data around the sample mean.

$$\sigma = \sqrt{\frac{\sum(x_i - \mu)^2}{N}}$$

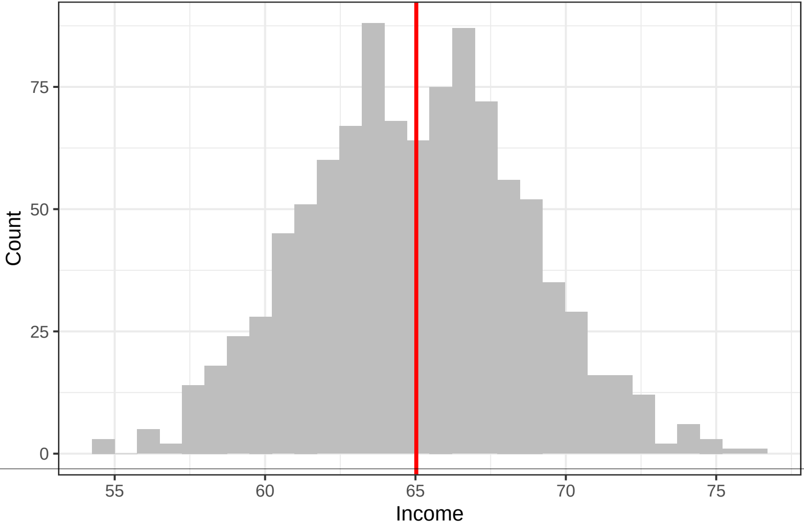
- **Standard error** = a measure of the spread of sample means around the population mean; the standard deviation of the sampling distribution.

Sampling distribution



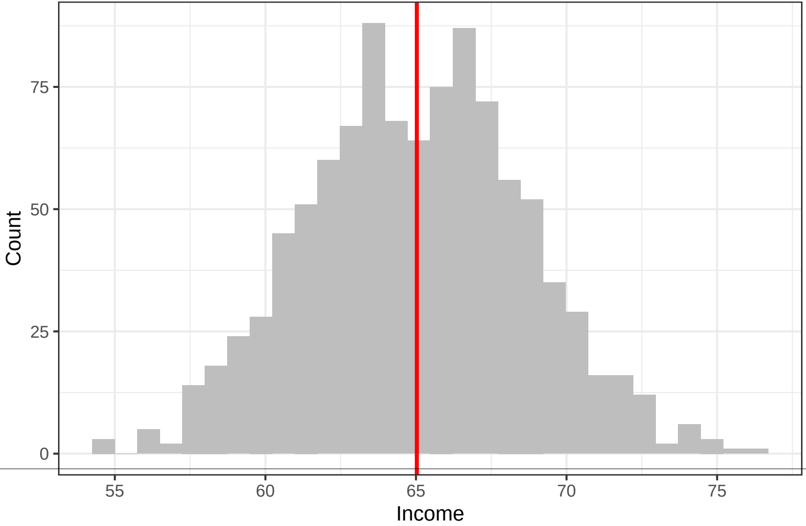
Sampling distribution

Sampling distribution of the mean:



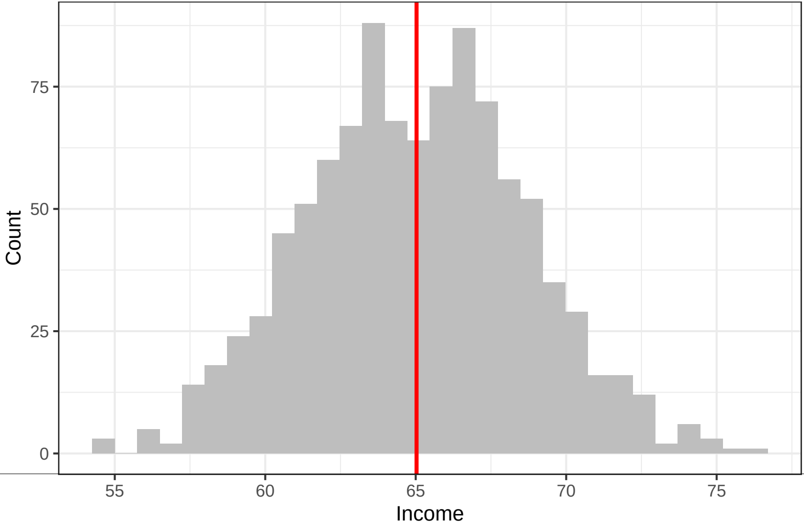
Sampling distribution

What do you notice about the shape of this distribution?

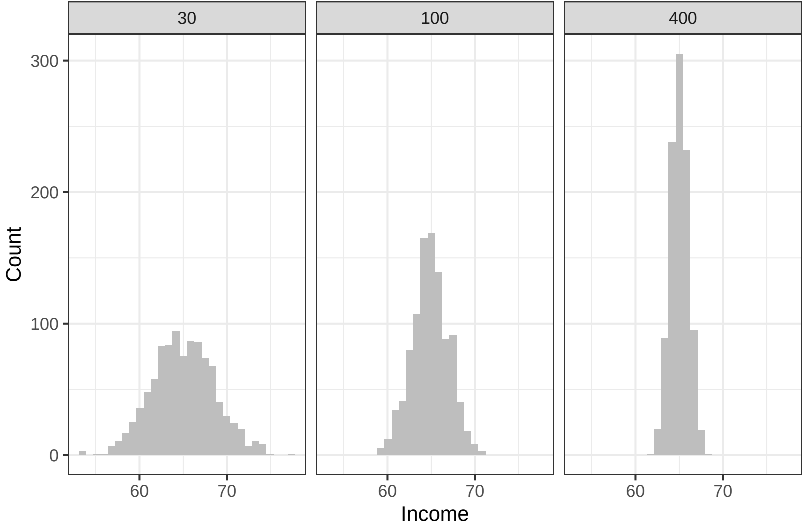


Sampling distribution

- **Central Limit Theorem**

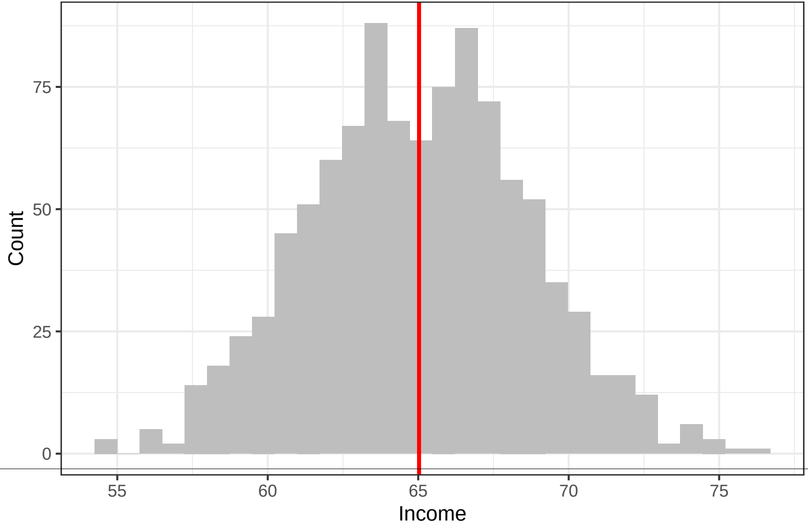


Sampling distribution



Sampling distribution

How do I interpret the standard deviation of this distribution?



Sampling distribution

- However defined, the standard error is a measure of the variation in an estimate and gets smaller as sample size gets larger, converging on zero as the sample increases in size.

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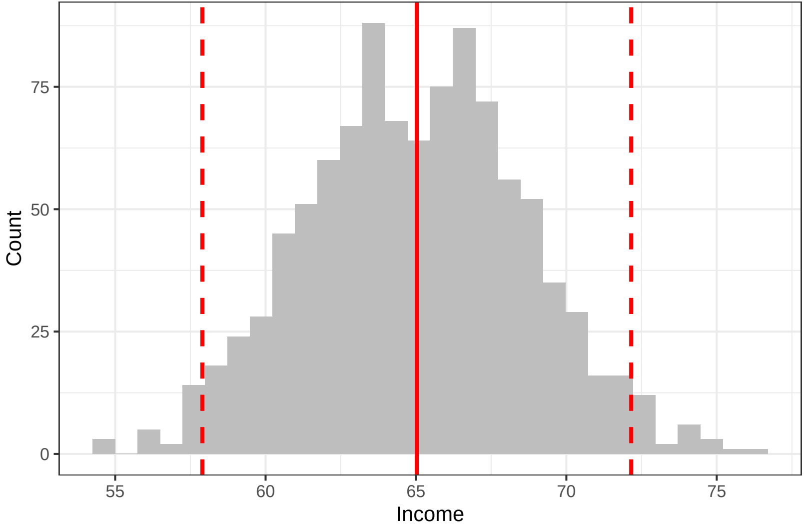
- However defined, the standard error is a measure of the variation in an estimate and gets smaller as sample size gets larger, converging on zero as the sample increases in size.

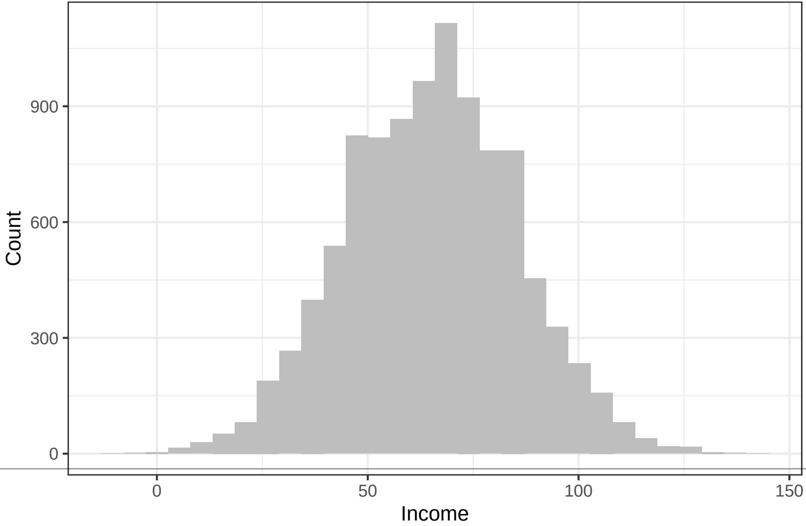
- The **confidence interval** represents a range of values of a parameter or quantity of interest that are roughly consistent with the data, given the assumed sampling distribution.

Sampling distribution



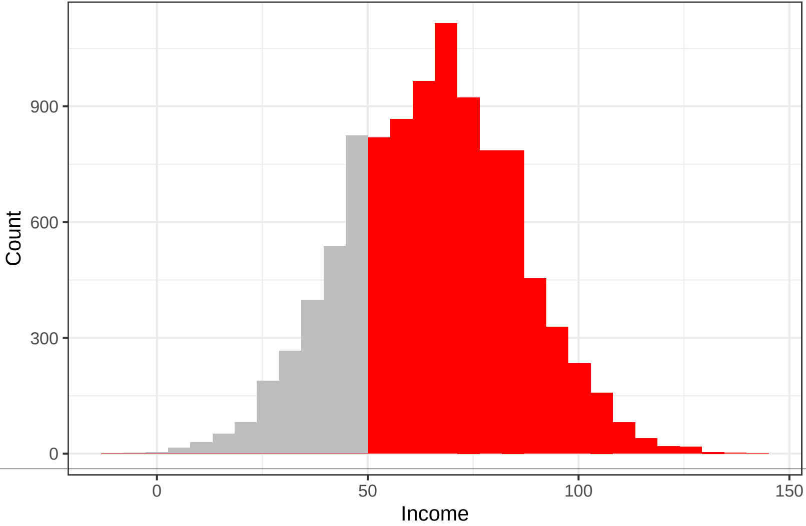
Bias

It is possible to be very confidently wrong!



Bias

Say only people with incomes above 50k complete my survey:



Describing uncertainty:

- Standard errors capture a particular (but very important!) type of uncertainty: sampling error.

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- Standard errors capture a particular (but very important!) type of uncertainty: sampling error.
- There are lots of other reasons to be uncertain!

Statistical significance

What can we do with standard errors?

- Hypothesis testing